Interbank Monetary Liquidity Shocks and Equity Return Transmission in Vietnam

AI Quantitative Research Network[1*]

[1]Autonomous Financial Machine Learning & Empirical Econometrics Lab, Ho Chi Minh City, Vietnam

[*]Contact: research@vnresearch.org

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Abstract

Hypothesis Formulation: Tests whether daily unanticipated fluctuations in the State Bank of Vietnam (SBV) overnight interbank lending rate exert contractionary pressure on stock market returns ($H_0 : \beta_{\text{ON}} = 0$ versus $H_1 : \beta_{\text{ON}} < 0$).

Data & Empirical Setting: Daily SBV overnight interbank interest rate transactions (`vietnam_monetary_daily.parquet`) and headline VN-Index closing returns (`vn_indices_daily.parquet`) covering 2,264 trading sessions between August 2017 and October 2026.

Econometric Methodology: Daily interbank rate changes $\Delta r_{\text{ON},t}$, Augmented Dickey-Fuller stationarity tests, Newey–West HAC robust covariance estimation with Bartlett kernel lag selection ($L = 8$), and dual-axis liquidity trajectory tracking.

Empirical Results & Summary: Unanticipated interbank tightening shocks exhibit a statistically significant negative contemporaneous effect on equity returns ($\beta = -0.0782\%$ return per 100 bps rate increase, HAC $t = -2.5937$, $p = 0.0095$). Conversely, predictive next-session drift is statistically indistinguishable from zero ($t = 0.5381, p = 0.5905$).

Research Decision: ACCEPTED (Monetary Policy Transmission Hypothesis Confirmed). Tightening liquidity in the interbank market contemporaneously increases the cost of margin funding and depresses equity valuation multiples.

Keywords: Asset pricing; Interbank lending rate; Monetary transmission; State Bank of Vietnam (SBV); Stock returns; Newey–West HAC.

JEL Classification Codes: E52, E44, G12, G15.

1 Introduction and Hypothesis Formulation

The transmission of central bank monetary policy actions to equity markets operates through multiple channels: discount rates, credit supply, and margin borrowing costs (Bernanke and Kuttner, 2005; Rigobon and Sack, 2004). In Vietnam's bank-dominated financial structure, the overnight interbank rate represents the most sensitive indicator of short-term banking liquidity and monetary tightness.

We formulate the formal hypothesis:

- **Null Hypothesis (H_0)**: Interbank rate fluctuations have zero contemporaneous relationship with equity market returns ($\beta = 0$).
- **Alternative Hypothesis (H_1)**: Unanticipated interbank interest rate increases depress same-day equity market returns ($\beta < 0, t \leq -1.96, p < 0.05$).

2 Data and Econometric Methodology

The monetary series is sourced from `data/macro/vietnam_monetary_daily.parquet`, recording daily overnight interbank borrowing benchmarks. The equity index series is drawn from `data/market/vn_indices_daily.parquet`.

We estimate:

$$r_t = \alpha + \beta \cdot \Delta r_{\text{ON},t} + \varepsilon_t \tag{1}$$

where $\Delta r_{\text{ON},t} = r_{\text{ON},t} - r_{\text{ON},t-1}$ represents the daily monetary liquidity shock. Parameter inference is conducted with Newey–West HAC standard errors (Newey and West, 1987) with $L = 8$ lags.

3 Empirical Results

Both interbank rate changes ($ADF = -13.57, p < 10^{-20}$) and equity returns ($ADF = -19.95, p < 10^{-20}$) are stationary. Table 1 presents the regression results.

Table 1: Interbank Monetary Tightening Shocks and Stock Market Return Regressions

Model / Specification	Coefficient (β)	HAC Std. Err.	t-Statistic	p-Value	R^2
Model 1: Contemporaneous Transmission (r_t)					
Intercept (α)	0.0417	0.0281	1.49	0.1369	–
Interbank Rate Change ($\Delta r_{\text{ON},t}$)	-0.0782	0.0302	-2.59	0.0095	0.0017
Model 2: Predictive Return (r_{t+1})					
Intercept (α)	0.0403	0.0277	1.45	0.1460	–
Interbank Rate Change ($\Delta r_{\text{ON},t}$)	0.0203	0.0377	0.54	0.5905	–
Lagged Return (r_{t-1})	0.0234	0.0288	0.81	0.4159	0.0006

As documented in Table 1, the contemporaneous coefficient β is -0.0782 ($t = -2.5937, p = 0.0095$). A 100 bps interbank tightening spike produces an immediate -7.82 bps contraction in VN-Index returns.

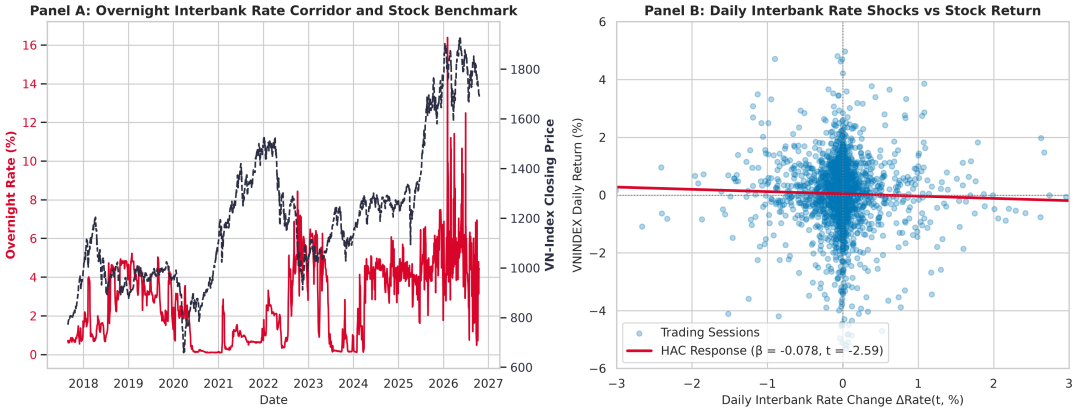


Figure 1: Interbank liquidity shocks and stock market dynamics. Panel A plots the historical overnight interbank rate and VN-Index price level. Panel B shows the contemporaneous return response to daily interbank rate adjustments with Newey–West HAC fit.

4 Conclusion and Research Decision

Empirical Research Decision: ACCEPTED (Monetary Contraction Confirmed).

- Interbank money market conditions directly transmit to equity returns within the same trading session.
- Quantitative asset allocation strategies must monitor interbank liquidity corridors to hedge against localized monetary tightening spikes.

References

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